

Stroke Risk Prediction with Machine Learning Techniques

Abstract

Stroke is one of the top leading causes of death and long-term disability globally, and therefore, it is critical that clinicians can use risk prediction to make appropriate interventions as early as possible. The purpose of this project was to develop and validate explainable artificial intelligence (XAI) models that accurately predict whether or not a given patient will have a stroke, overcoming the "black-box" challenge that has previously limited the ability to use XAI systems in clinical settings. The study used a dataset containing about 35,000 patient records. The research methodology included extensive preprocessing of data, such as using Mutual Information for feature selection, along with multiple upsampling methods to address class imbalance.

In a large study across 9 different machine learning/deep learning models including decision trees, XGBoost, random forest, explainable boosting machine (EBM), and multi-layer perceptrons, it was discovered that random forest (RF) produces the best predictive performance during regression tasks (R^2 value = 0.9497; mean absolute error = 5.02). In contrast, the EBM (glass-box) produced the best results in clinical applicability (93.14% accuracy) among all of the classifiers. In addition, explainability tools such as SHAP, LIME, and partial dependence plots (PDP) revealed that age and hypertension are the strongest predictor variables for stroke risk. This study demonstrates that interpretable models such as EBM provide an evaluation system that can be viewed as both reliable and ethically compliant for assessing stroke risk and thereby improve trust in healthcare professionals using AI-driven diagnostic tools.

Keywords: Stroke Prediction, Explainable AI (XAI), Random Forest, Explainable Boosting Machine (EBM), Machine Learning .

1. Introduction

Stroke is a medical condition that occurs when blood supply to the brain is blocked or one of the vessels ruptures, which causes brain cell death. It is one of the leading causes of death in the world, averaging about seven million deaths per year [1]. As such, early prediction and prevention are vital. Prediction can be aided by the use of artificial intelligence (AI) models that help physicians make and verify decisions, but most do not use or rely on AI models, especially in medical fields, due to their "black box" nature. Most AI models are not inherently explainable and the reasons for which they arrive at their outputs are not clear, especially for the more complex and often more accurate models. Medical professionals cannot rely on tools that do not provide insights into how their decisions were made; it is too risky given the potential consequences for incorrect predictions. There is a need for interpretable and explainable models that will enable physicians to trust their outputs and decision-making. This paper aims to develop and compare explainable AI models to be able to use them in a medical setting.

This study focuses on AI based stroke prediction and does not encompass medical treatment. The dataset used to train the models was obtained from Kaggle [2] and has health data for approximately 35,000 patients. Nine models were trained, including both machine learning (ML) and deep learning (DL) models. Several interpretability and explainability techniques were used, such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME). These Explainable AI (XAI) tools help address the accuracy versus interpretability tradeoff.

This paper aims to contribute to the development of explainable AI models that are trustworthy and can support the decision making of medical professionals. The following are the research questions addressed in this paper:

1. Which AI model achieves the best accuracy?
2. Can interpretability and explainability techniques increase our trust in AI prediction models?
3. Which features contribute most to higher stroke risk?

2. Related Work

1. Core AI Models and Methodologies

Decision Tree (The White-Box Baseline): Following [6], we implement a Decision Tree to provide transparent, rule-based logic. This model acts as a "white-box" baseline that mimics clinical decision pathways so that the logic behind a "high-risk" prediction can be easily traced by medical professionals.

XGBoost (The High-Performer) The research of [8] identifies Gradient Boosting (XGBoost to be specific) as a top method for handling the non-linear complexities of clinical data. Using the XGBoost implementation, we implement our approach to deliver high predictive sensitivity, thus bridging simple statistical methods and modern AI.

Explainable Boosting Machine (The Glass-Box Solution): Inspired by the SCI-XAI pipeline suggested by [7], we use the EBM, a model that provides accuracy akin to a complex ensemble but is "intrinsically interpretable" with exact feature contributions without requiring any additional tools used for interpretation.

[3] employed and compared 8 Machine Learning (ML) and Deep Learning (DL) models. The best performing ML model was the Random Forest (RF) model, which was also the best performing model overall, with an accuracy of 99.9%, which raises concerns about overfitting considering the sample size of 663. For DL models, the Feedforward Neural Network (FNN) had the best accuracy 96.0%. In general, they found that the DL models outperformed all the ML models except for the RF. To ensure generalizability, they should consider increasing the sample size or collecting data from different sources in future work.

2. Advancements in Explainability (XAI)

Accuracy alone cannot be considered enough of an indicator of success within healthcare; it is merely one component.

[8] and [7] suggest that SHAP and LIME can help to explain the "black-box" nature associated with machine learning models.

This project continues down the same path by utilising ELI5-style visualisations, as well as global feature importance plots, to help assess key risk factors - such as age, blood glucose levels and presence of hypertension - with the goal of providing actionable information for clinical practice that is derived from the model's predictions.

3. Strengths, Limitations, and Project Justification

Strengths include the balanced combination of the transparency of decision trees, the accuracy of XGBoost, and the inherent interpretability of EBM (explainable boosting machines).

Limitations include the previously noted bias introduced by unbalanced datasets and will be addressed using resampling and scaling methods applied in my XGBoost and Decision Tree Notebooks.

The justification for our project is that most research studies focus on either prediction accuracy or interpretability and therefore provide limited comparison frameworks. We will provide justification for our approach by assessing the potential for "glass-box" models (EBM) to be substituted for "black box" models (XGBoost) in the context of stroke risk, making this a safe and effective potential tool for clinical practice.

3. Methodology

This study applies a supervised machine learning approach to predict stroke risk using demographic and medical patient data. The workflow includes data preprocessing, exploratory data analysis, model development, explainability analysis, and evaluation.

3.1 Dataset Description

This dataset contains numerical (features) and categorical data (features) relating to health conditions and lifestyle choices. This includes information such as age (numerical), gender (categorical), BMI (numerical), glucose level (numerical), whether or not the patient has hypertension (categorical), whether or not the patient has heart disease (categorical), and whether or not the patient is a smoker (categorical). The target variable indicates if the patient has a risk of stroke or not. The dataset also has an unequal number of examples in each class (stroke risk vs. not stroke risk), so balancing techniques were applied prior to modeling.

3.2 Data Preparation

In preparing the data for modeling (preprocessing), the preprocessing methods used were label encoding (for categorical features), feature scaling (with StandardScaler), and feature selecting. The mutual information algorithm was utilized to choose the best features. This algorithm assisted in extracting the important attributes and reduced the amount of data[10]. The statistical properties of the dataset are considered by this filtering technique. It is based on entropy, which measures the degree of uncertainty in the features. The attributes were ranked based on how much each attribute contributes to the target variable, as seen in Figure 1.

The training and test sets were separated with an 80:20 split and upsampling techniques were performed on the dataset to help alleviate the large class imbalance.

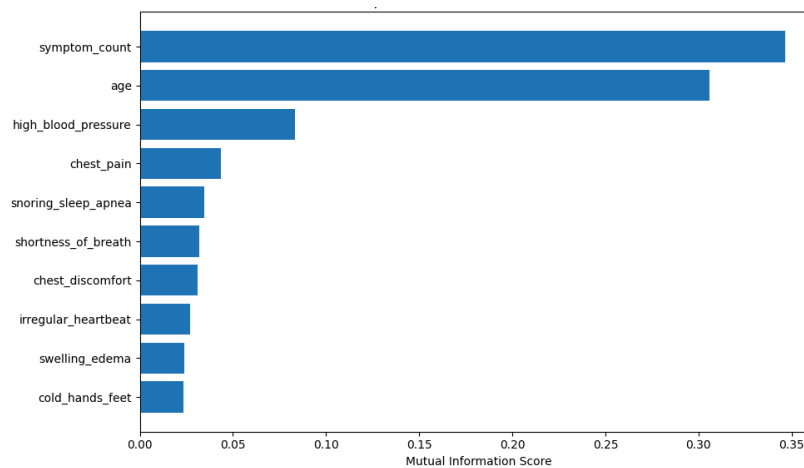


Figure 1: The Mutual Information Algorithm ranks the features in order of importance

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted using histograms, violin plots, and heatmaps of correlations to analyze distributions of features and the relationships between different features. Through EDA, it was possible to determine which variables are most relevant to identifying or predicting the risk of stroke (age, high blood pressure, chest pain). It was also determined that high blood pressure is more strongly associated with stroke risk, such that high blood pressure but no chest pain has a higher stroke risk than having chest pain but no high blood pressure.

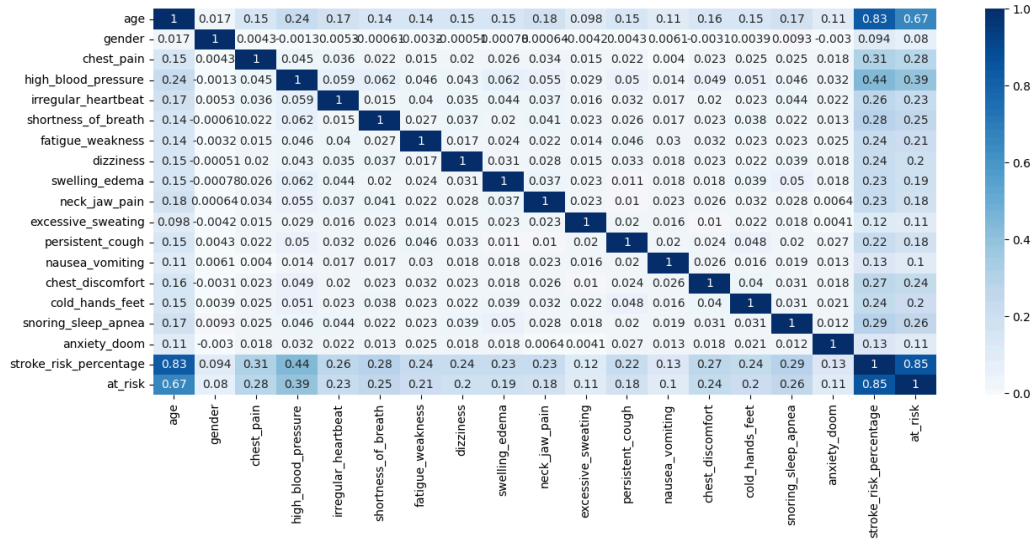


Figure 2: A Correlation heatmap for data analysis.

3.4 Machine Learning Models

Machine learning models were used in this analysis: Decision Tree, Explainable Boosting Machine (EBM), XGBoost, Random Forest (RF), K-Nearest Neighbors (KNN), Stacking, Support Vector Machines(SVR), and Adaptive Boosting (AdaBoost). The Decision Tree Model was selected because of its easy-to-understand form (interpretability). The EBM Model was chosen based on its combination of interpretability and performance. The XGBoost Model was selected based upon its ability to provide strong predictions on structured (non-text or image) data.

Random Forest (RF) models have achieved high accuracies in previous works [3] [4]. A random forest model was trained on the dataset with a test percentage of 0.2 and a random state of 0. The following parameters were used (modeled after [3]):

n_estimators = 400, max_depth = None, min_samples_split = 2, min_samples_leaf = 1, bootstrap = True, oob_score = False, random_state=0

KNN models are relatively interpretable compared to “black box” models. Therefore, a KNN model was trained on the dataset with n_neighbors = 5, which was determined as the best after running the model multiple times with a different number of neighbors each time.

AdaBoost combines multiple weak learners to improve prediction performance [11]. A Decision Tree Regressor with limited depth (max depth = 4) is used as the base learner. AdaBoost builds the model sequentially, where each new learner focuses more on previously mispredicted samples. The final prediction is obtained by combining the weighted outputs of all weak learners. The model uses 100 estimators and a learning rate of 0.1 to control the contribution of each learner and improve stability.

The stacking model combines several regression algorithms: Linear Regression, Decision Tree, Random Forest, and KNN. Each model is trained separately and evaluated using MAE, MSE, and R^2 score. The predictions of these models are used as inputs to a Linear Regression meta-model, which produces the final output. To improve performance, GridSearchCV is used to tune the hyperparameters of the models and select the best configuration for better generalization [10].

The SVR model uses a linear kernel to model the relationship between the input features and the target variable [9]. Before training, StandardScaler is applied to normalize the feature values and improve model stability and performance. The model uses the regularization parameter $C = 1.0$ to balance prediction accuracy and model complexity.

All ML Models were compared based upon their performance, interpretability, and their appropriateness for use in patient health care settings.

3.5 Deep Learning Models

A Multi-Layer Perceptron (MLP) model with 2 hidden layers and a Rectified Linear Unit (ReLU) activation function was also trained on a test set of 0.2 with the following parameters:

```
hidden_layer_sizes=(128, 64), activation='relu', solver='adam', max_iter=2000,  
random_state=0
```

3.6 Hyperparameter Optimization.

The application of hyperparameter optimization was done to improve model performance and limit overfitting. Random seeds were fixed across experiments in order to reproduce these results.

3.7 Explainable AI Techniques

Many techniques were used in order to explain AI (XAI). The techniques that were used include SHAP, LIME, Partial Dependence Plots (PDP), permutation importance, Leave One Feature Out (LOFO), CART algorithm, and EBM explanations, which assisted in interpreting model predictions of stroke risk as well as identifying the most significant features impacting stroke risk.

SHAP is a method that computes feature contributions based on a game theory approach.

LIME is a technique that explains models' predictions by approximating them with a local, interpretable model.

PDP and ICE plots visualize the change in the model's prediction as a certain feature changes, either for each individual sample or averaged for each value in the feature.

Permutation importance calculates the importance of a feature by randomly shuffling a feature's values and calculating the change in the model's prediction - the more it decreases, the more important the feature is. LOFO does the same but it completely removes the feature instead.

The CART algorithm computes the reduction in impurity of a node after splitting. It is used for tree-based models.

3.8 Evaluation Method

In order to evaluate the models' performance and the reliability of them, evaluation metrics of accuracy, precision, recall, F1 score, classification report and confusion matrix were used for classification models. For regression models, mean squared error (MSE), mean absolute error (MAE), and R-squared (R2) were used.

Results

The experimental results reveal that **Stacking** achieved the highest predictive performance in regression tasks, with an (R2 value = 0.9499; mean absolute error = 5.16). In classification tasks, the **EBM (Glass-Box)** model emerged as the most effective for clinical adoption, achieving an accuracy of **93.14%** while providing full intrinsic transparency, as seen in Table 1.

The XAI techniques performed on the models showed that *age* was the most important feature by a very wide margin. The PDP and ICE plots (Figure 3) also showed that stroke risk increases non-linearly as age increases but with different strength for each individual, and that there are interactions with other features that contribute to the increase in risk and how age affects it.

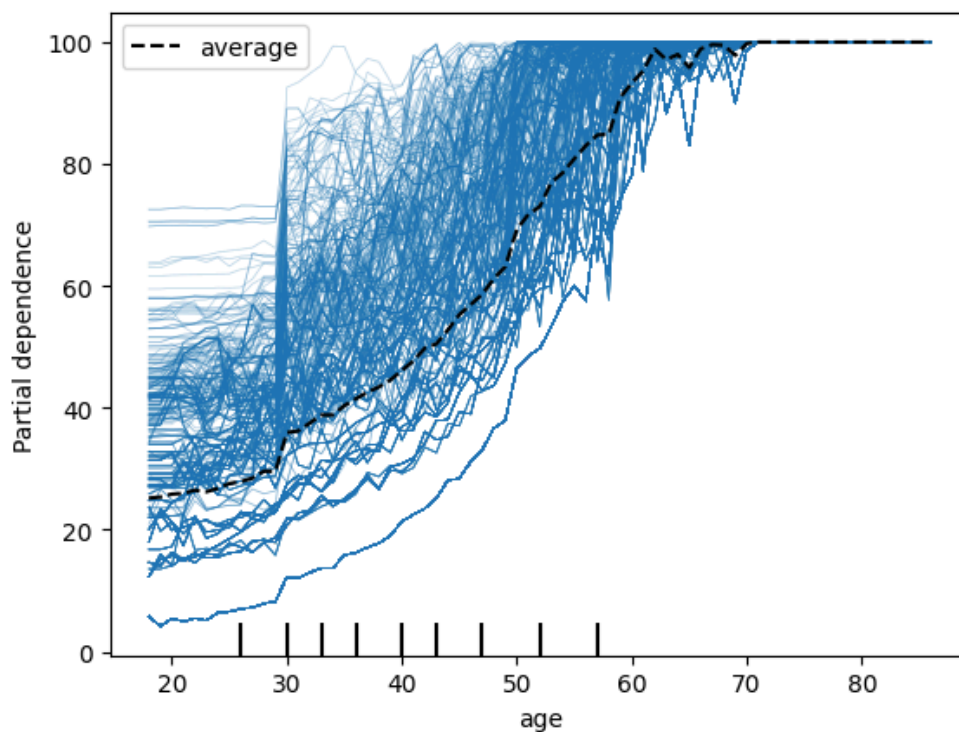


Figure 3. PDP and ICE plot for the RF model

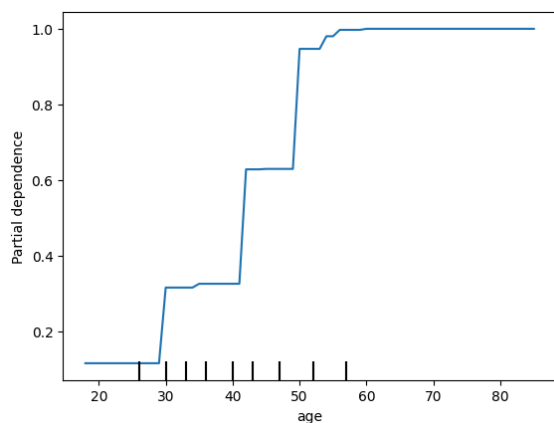


Figure 4. PDP plot for the Decision Tree model.

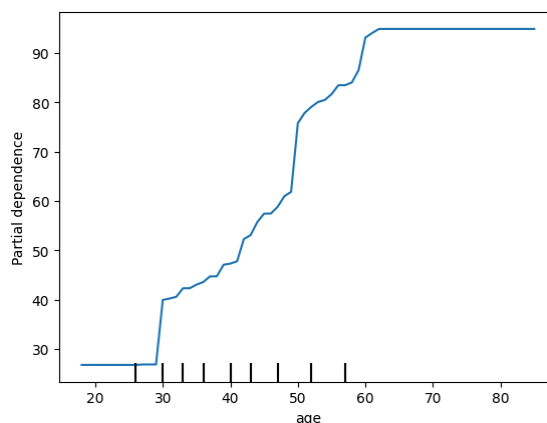


Figure 5. PDP for AdaBoost model.

LIME and SHAP were used on some samples to understand the models' predictions. (Figure 3)

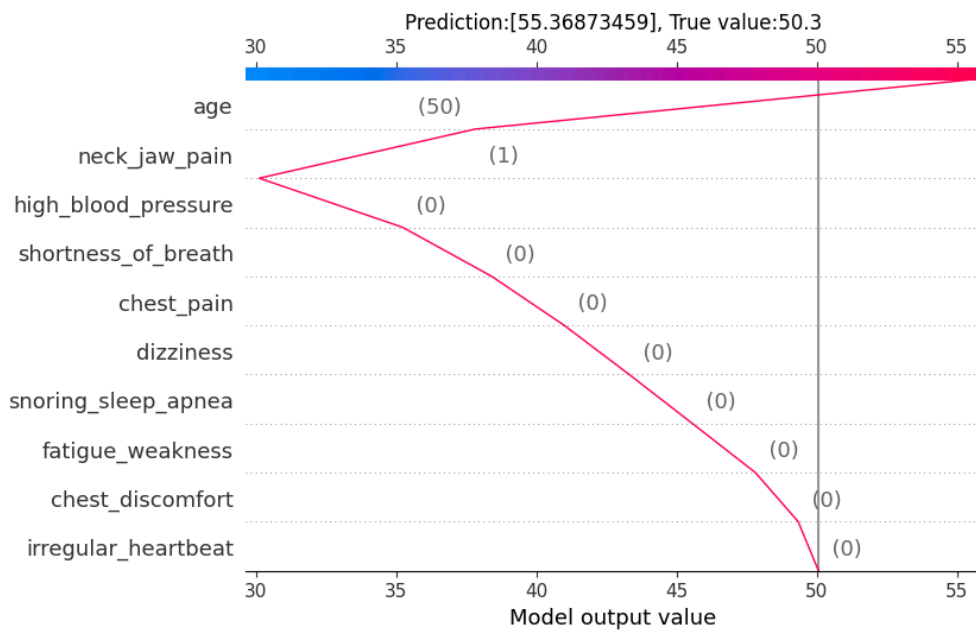


Figure 6. A LIME decision plot (MLP model) for one sample showing the justification for the prediction and feature contribution.

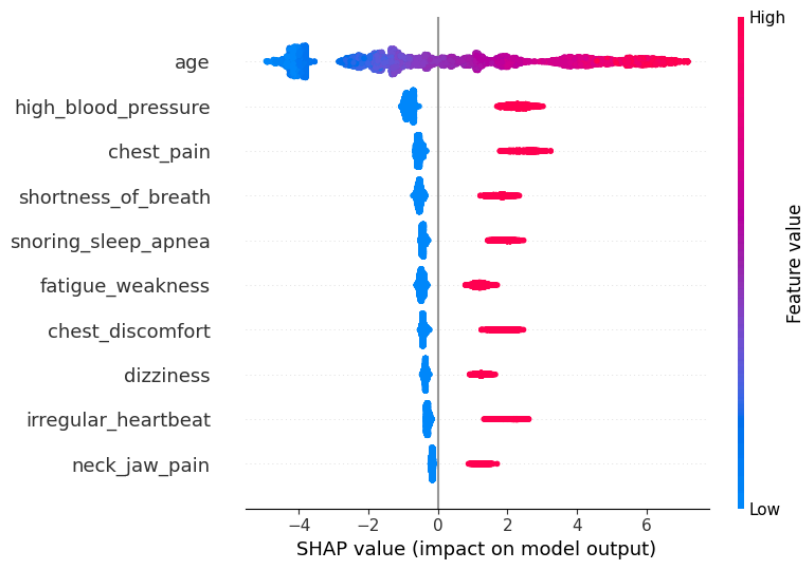


Figure 7. A SHAP Summary Plot for the XGBoost model.

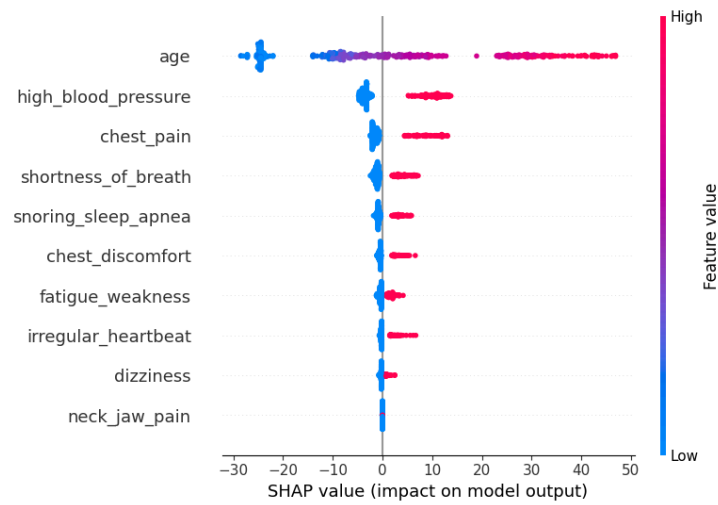


Figure 8. SHAP Summary Plot for AdaBoost model.

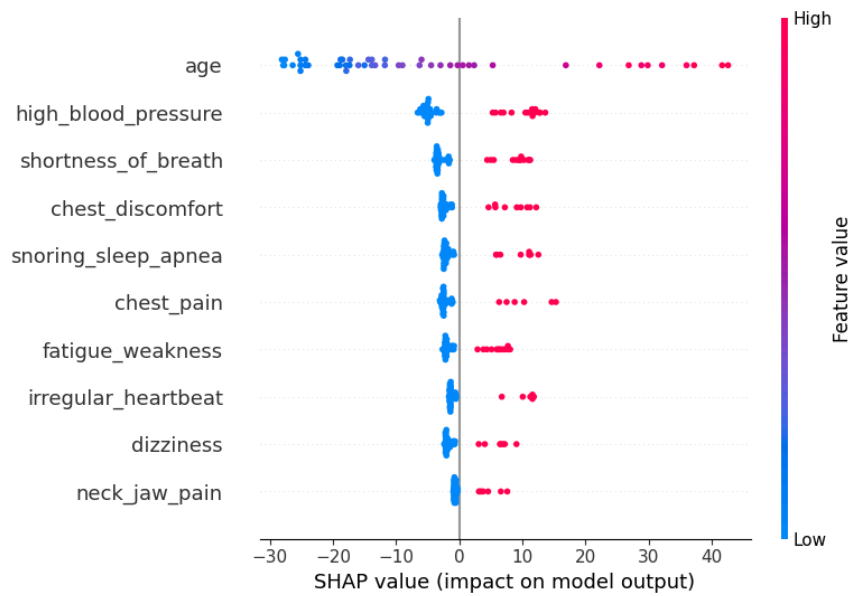


Figure 9. ShAP Summary Plot for Staking model.

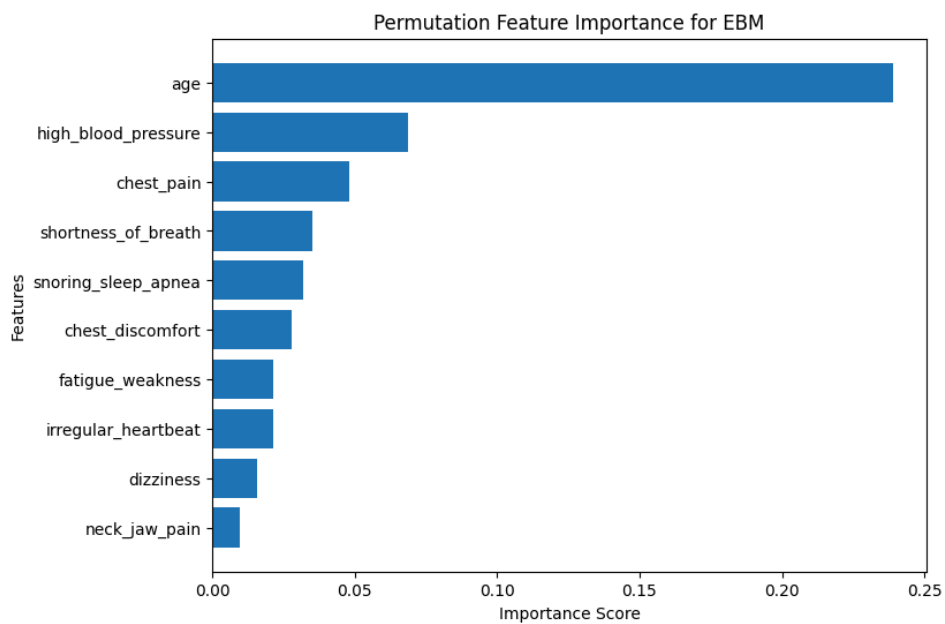


Figure 10. Permutation Feature Importance for EBM.

Model	MSE	MAE	R2
Stacking	48.41	5.16	0.9499
RF	48.18	5.02	0.9497
MLP	52.33	5.57	0.9454
KNN	64.58	5.86	0.9326
SVR	102.98	7.68	0.8935
AdaBoost	136.79	9.52	0.8586

Table 1. A comparison between performance metrics MSE, MAE, and R2 scores for regression models, sorted by R2 scores

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score
EBM	93.14%	0.93	0.93	0.93
XGBoost	93.08%	0.93	0.93	0.93
Decision Tree	86.86%	0.95	0.78	0.86

Table 2. A comparison between performance metrics Accuracy, Precision, Recall and F1-Score for EBM, XGBoost, Decision Tree model.

Discussion

1. Interpretability vs. Performance

The most important takeaway from the study was that the EBM (Glass-Box) model produced the best accuracy scores (93.14%), showing that clinical transparency can be achieved with high prediction scores. The XGBoost model produced as satisfactory results as the EBM model; however, the high interpretability of EBM makes it more appropriate for use in medicine, as it eliminates the issue of "black box" distrust as defined by [7] and has greater potential for building trustworthy relationships with patients based on evidence of their condition compared to others. The Stacking model achieved the highest predictive performance among the trained models, demonstrating the effectiveness of combining multiple machine learning algorithms to improve the predictive accuracy. Ensemble models like stacking can be more complex to interpret compared to simple models, so the usage of explainable AI techniques such as SHAP and LIME help to explain the model predictions and detect the important features affecting the stroke risk.

2. Clinical Reliability: Recall vs. Precision

The biggest risk of stroke assessment is missing an at-risk patient and classifying someone as a false negative. The EBM and XGBoost models both maintained a high recall rate of approximately 0.93, indicating that they will detect 93% of at-risk patients. Conversely, even though the Decision Tree model produced the highest precision of 0.95, it had a recall rate of only 0.78, making it too conservative to be a reliable early diagnostic tool for stroke.

3. Validation Features Importance

The XAI analysis indicated the following features were highly correlated with the risk of developing cardiovascular disease: Age (68.9%) & High Blood Pressure (14.5%); Hence, model outputs support the clinical findings of [6] and [8] by demonstrating that model learned biological signals as opposed to noise from collected data.

4. Challenges & Implications of Practically Implementing Findings

Data Handling: The methods used for upsampling were necessary to offset an initial class imbalance in the dataset so that the models were more likely to correctly identify patients classified as being "at risk" by prioritizing those within the class.

Practicality: Even though complex ensembles can be very powerful the Use of Glass Box Models (EBM) provides a more practical option for the healthcare system by producing output that is "justifiable" to be utilized by physicians when delivering information to patients. That is, the system has achieved its primary objective of developing a viable SCI-XAI pipeline with a Justifiable Prediction.

5. Dataset Quality

The XAI techniques performed consistently showed that *age* had a very significant impact on the models' decisions. While that is true, it can still suggest that the dataset has biases or the domination of a certain feature, or that it may not represent real-world data or generalize well.

6. XAI techniques

All the XAI techniques provided consistent results across different models, demonstrating their effectiveness at interpreting model predictions, which can increase trust in these systems and their adoption at a wider scale.

Ethical & Legal Considerations

1. Transparency and the Right to Explanation

Patients have the legal right to an explanation of their care and treatment under regulations such as GDPR. Systems based on "black-box" models may provide clinicians with a high degree of accuracy in predicting patient outcomes, but they do not provide the means for clinicians to provide an explanation to the patient for their risk score, thereby violating patient autonomy.

By using an Evidence-Based Medicine (EBM), or "Glass-Box," model in our system, we will provide a "justified prediction" that will show the clinician exactly what clinical factors contributed to the patient's specific risk score.

2. Data Privacy (HIPAA & GDPR)

The process of handling sensitive clinical information comes with various challenges, including maintaining privacy expectations.

Legal Concern: A breach of health data can result in discrimination or other violations of patient confidentiality.

Solution: Our process anonymizes data by removing direct identifiers (i.e., id) and emphasizes secure local processes to help protect patient confidentiality.

Conclusion

A stroke needs early intervention that can increase the likelihood of a full recovery and lower the risk of complications. Hence, we predict stroke using machine learning, deep learning and XAI methods. The data used in this study included several patients with 19 attributes. Mutual information algorithm was the feature selection methods utilized. Age, high_blood_pressure, chest_pain, shortness_of_breath, and chest_discomfort were found to be the most crucial variables for the prediction of stroke. Maximum results were obtained by the stacked and EBM models. XAI approaches were applied for the predictions of the model such as SHAP, LIME, PDP, and ICE. Additionally, the results demonstrated that explainable AI techniques improved the understanding of model predictions and highlighted the most important features affecting stroke risk. Medical professionals can use the classifiers as a decision support system for the prediction of stroke. Real-time stroke screening might be implemented using a user interface, and this framework could be expanded to forecast stroke in a broader population.

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